BTS–VI(R/S)–05–24–2480 Reg. No. | | | | | | | | |

**B**

# *B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024*

## ME 19-205-0606 (IE) ADDITIVE MANUFACTURING

*(2019 Scheme)*

Time: 3 Hours Maximum Marks: 60

Course Outcome
On successful completion of the course, the students will be able to:

CO1: Understand the concept of AM.
CO2: Study various AM technologies using the basics of computer numerical control.
CO3: Study the modeling of AM processes.
CO4: Study the application of various fields.

Bloom`s Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

### PART A
(Answer *ALL* questions)

| | | (8 × 3 = 24) | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| I. | (a) | Define a prototype and enumerate its roles. | 3 | L1 | 1 | 1 |
| | (b) | What is additive manufacturing? List out its advantages. | 3 | L2 | 1 | 1,2 |
| | (c) | "Anisotropy is an important problem for AM".Explain. | 3 | L2 | 2 | 1,2 |
| | (d) | Explain about functionally graded materials. | 3 | L1 | 2 | 1,2 |
| | (e) | What are the differences between selective laser sintering and selective laser melting? | 3 | L1 | 4 | 1,2 |
| | (f) | What are the implications of vacuum in electron beam powder bed fusion? | 3 | L2 | 4 | 1,2 |
| | (g) | Enumerate various numerical methods for deposition modeling. | 3 | L1 | 3 | 1,2 |
| | (h) | In the parlance of modelling of free surface flow how, do you distinguish between Lagrangian methods and Eulerian methods. | 3 | L2 | 3 | 1,2 |

### PART B

(4 × 12 = 48)

| | | | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| II. | (a) | Explain the Generic AM Process. | 7 | L2 | 1 | 1,2,3 |
| | (b) | What is indirect Rapid Tooling? | 5 | L2 | 1 | 1,2 |
| | | **OR** | | | | |
| III. | (a) | Explain about AM application levels with sketches. | 7 | L1 | 1 | 1,2 |
| | (b) | Explain the basic theory of reverse engineering. | 5 | L2 | 1 | 1 |

**(P.T.O.)**

| | | | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| IV. | (a) | Discuss about various materials used in AM. | 7 | L1 | 2 | 1,2, |
| | (b) | Compare the anisotropic behaviour of materials in various AM technologies. | 5 | L2 | 2 | 1,2, |
| | | **OR** | | | | |
| V. | (a) | Explain about various factors affecting solidification in AM. | 7 | L2 | 2 | 1,2, |
| | (b) | What kinds of plastics can be processed by AM? Plot them in the plastic triangle. | 5 | L1 | 2 | 1,2, |
| VI. | (a) | Explain laser powder bed fusion process with a neat sketch. | 6 | L1 | 4 | 1,2, |
| | (b) | Explain various components needed to realize electron beam powder bed fusion. | 6 | L1 | 4 | 1,2, |
| | | **OR** | | | | |
| VII. | (a) | Explain the working principles of fused deposition modeling. | 7 | L1 | 4 | 1,2, |
| | (b) | Explain about process planning in AM.. | 5 | L1 | 4 | 1,2, |
| VIII. | | Explain the governing equations used in the mathematical modeling of laser metal deposition process | 12 | L2 | 4 | 1,2. |
| | | **OR** | | | | |
| IX. | | Explain thermo-hydrodynamic simulation in Additive Manufacturing Processes. | 12 | L2 | 4 | 1,2, |

Blooms's Taxonomy Level's
L1 – 55%, L2 – 45%.

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